DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1
26MX16 - PROGRAMMING AND DATA STRUCTURES LABORATORY
PRACTICE SHEET 6

Prerequisite	C fundamentals, conditional statements, loops, 1D arrays, strings, array traversal, searching and 2D arrays.

1. An IT company records the total hours a server was available and the total monitored hours. Calculate the server uptime percentage.
   Function Prototype
   float calculateUptime(int availableHours, int totalHours);
   Business Rule
   Uptime % = (Available Hours / Total Hours) × 100
   Sample Input
   95 100
   Sample Output
   Uptime: 95.00%

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2. A data center records CPU loads of three servers. Find the server having the highest CPU load.
   Function Prototype
   int findHighest(int a, int b, int c);
   Sample Input
   72 91 85
   Sample Output
   Highest Server Load: 91

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3. An API monitoring system records response time in milliseconds. Write a function that returns the category of the response.
   Function Prototype
   int classifyResponse(int responseTime);
   Business Rules

Response Time	**Return Value**
≤ 200 ms	1
201–500 ms	2
> 500 ms	3

   The main() function should display:
   1 → FAST
   2 → ACCEPTABLE
   3 → SLOW
   Sample Input
   650
   Sample Output
   Response: SLOW

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4. An IT company gives employees a bonus based on their performance score.
   Function Prototype
   float calculateBonus(float salary, int score);
   Business Rules

Score	**Bonus**
90–100	20% of salary
75–89	15%
60–74	10%
<60	5%

Sample Input
50000 85
Sample Output
Bonus: 7500.00

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5. A cybersecurity system stores login results in an integer array:
   - 1 → successful login
   - 0 → failed login

   Write a function to count successful logins.
   Function Prototype
   int countSuccessful(int login[], int n);
   Sample Input
   7
   1 0 1 1 0 0 1
   Sample Output
   Successful Logins: 4

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6. An employee ID list is stored in an array. Search for a given employee ID and return its first position**.**
   Function Prototype
   int findEmployee(int id[], int n, int key);
   Business Rule
   Return:
   - position of first occurrence if found
   - -1 if not found

   Sample Input
   7
   101 105 108 105 112 115 120
   105
   Sample Output
   First Occurrence: Position 2
   Positions start from 1**.**

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7. A web server maintains requests in a queue represented by an array. Rotate the requests to the right by K positions**.**
   Function Prototype
   void rotateRight(int arr[], int n, int k);
   Sample Input
   6
   10 20 30 40 50 60
   2
   Sample Output
   60 10 20 30 40 50
   Business Rule:
   If K > N, reduce it using:
   K = K % N

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8. A software monitoring system records daily performance scores. Find the length of the longest continuous increasing sequence.
   Function Prototype
   int longestIncreasing(int arr[], int n);
   Sample Input
   8
   40 45 50 42 48 52 55 30
   Sample Output

Longest Increasing Run: 4
The longest run is:
42 48 52 55

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9. An IT company wants to validate employee usernames. Write a function that returns 1 if the username is valid and 0 otherwise.
Function Prototype
int isValidUsername(char username[]);
Validation Rules:
- Length must be between 6 and 15.
- First character must be an alphabet.
- Only alphabets, digits and _ are allowed.
- Spaces are not allowed.

Sample Input
arun_2026
Sample Output
VALID
Sample Input
2arun@12
Sample Output
INVALID

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10. A cloud company stores CPU utilization of its servers in a 2D array. Write a function to calculate the total load of a specified server/team row**.**
Function Prototype
int rowTotal(int load[][10], int rows, int cols, int row);
Sample Input
3 4
40 50 60 70
30 45 55 65
80 70 60 90
2
Sample Output
Team 2 Total Load: 195
Then modify the program by writing another function:
int highestRow(int load[][10], int rows, int cols);
to identify the team having the highest total load.
Expected Output
Highest Load Team: Team 3

Skills Acquired	User-defined functions, function prototypes, parameter passing, return values, modular problem solving, array and string processing, array rotation and 2D array processing.